

Numerical Examples Line Codes

Example 1:

A random user binary stream employs polar RZ code ($\pm V$) with 50% duty cycle for transmission over $[0, T]$ secs for every symbol. Assuming the probability

$$P(+V) = 0.7$$

obtain the power spectral density and comment on the DC content in the waveform and how timing recovery at the receiver can be managed.

Solution:

Polar RZ (50%) duty cycle

$$\begin{aligned} g(t) &= \begin{cases} +V & \text{over } [0, T/2] \\ 0 & \text{over } [T/2, T] \end{cases} \\ G(f) &= \int_0^{T/2} V e^{-j2\pi ft} dt \\ &= V \left[\frac{e^{-j2\pi ft}}{-j2\pi f} \right]_0^{T/2} \\ &= \frac{V}{-j2\pi f} (e^{-j\pi fT} - 1) \\ &= \frac{V}{j2\pi f} (1 - e^{-j\pi fT}) \end{aligned}$$

Using

$$\begin{aligned} 1 - e^{-j\theta} &= 2je^{-j\theta/2} \sin\left(\frac{\theta}{2}\right) \\ G(f) &= \frac{V}{j2\pi f} \left[2je^{-j\pi fT/2} \sin\left(\frac{\pi fT}{2}\right) \right] \\ &= \frac{V}{\pi f} e^{-j\pi fT/2} \sin\left(\frac{\pi fT}{2}\right) \\ &= \frac{VT}{2} \text{sinc}\left(\frac{fT}{2}\right) e^{-j\pi fT/2} \end{aligned}$$

Therefore,

$$|G(f)|^2 = \frac{V^2 T^2}{4} \text{sinc}^2\left(\frac{fT}{2}\right)$$

Given,

$$P(+V) = 0.7$$

$$P(-V) = 0.3$$

$$\begin{matrix} (+1) & & (-1) \end{matrix}$$

For $k = 0$,

$$\begin{aligned} R(0) &= E[a_n^2] \\ &= 0.7(1)^2 + 0.3(-1)^2 \\ &= 0.7 + 0.3 \end{aligned}$$

$$= 1$$

For $k \neq 0$,

$$\begin{aligned} R(k) &= E[a_n]E[a_{n+k}] \\ &= (1)(1)(0.7)(0.7) + (-1)(-1)(0.3)(0.3) \\ &\quad + (1)(-1)(0.7)(0.3) + (-1)(1)(0.3)(0.7) \\ &= (0.49 + 0.09 - 0.21 - 0.21) \\ &= 0.16 \end{aligned}$$

Hence,

$$R(k) = \begin{cases} 1, & k = 0 \\ 0.16, & k \neq 0 \end{cases}$$

Now,

$$S_x(f) = \frac{|G(f)|^2}{T} \sum_{k=-\infty}^{\infty} R(k)e^{-j2\pi f k T}$$

Substituting,

$$S_x(f) = \frac{V^2 T^2}{4T} \text{sinc}^2\left(\frac{fT}{2}\right) \sum_{k=-\infty}^{\infty} R(k)e^{-j2\pi f k T}$$

Separating the $k = 0$ term,

$$S_x(f) = \frac{V^2 T^2}{4} \text{sinc}^2\left(\frac{fT}{2}\right) \left(1 + \sum_{\substack{k=-\infty \\ k \neq 0}}^{\infty} 0.16e^{-j2\pi f k T}\right)$$

Since

$$1 = 0.84 + 0.16$$

$$S_x(f) = \frac{V^2 T}{4} \text{sinc}^2\left(\frac{fT}{2}\right) \left(0.84 + \sum_{k=-\infty}^{\infty} 0.16 e^{-j2\pi f k T}\right)$$

Using

$$\sum_{k=-\infty}^{\infty} e^{-j2\pi f k T} = \frac{1}{T} \sum_{n=-\infty}^{\infty} \delta\left(f - \frac{n}{T}\right)$$

we get

$$S_x(f) = \frac{V^2 T}{4} \text{sinc}^2\left(\frac{fT}{2}\right) \left[0.84 + \frac{0.16}{T} \sum_{n=-\infty}^{\infty} \delta\left(f - \frac{n}{T}\right)\right]$$

Hence,

$$\boxed{S_x(f) = \frac{V^2 T}{4} \text{sinc}^2\left(\frac{fT}{2}\right) \left[0.84 + \frac{0.16}{T} \sum_{n=-\infty}^{\infty} \delta\left(f - \frac{n}{T}\right)\right]}$$

Has a strong DC component since

$$\text{sinc}^2(0) = 1$$

and

$$0.84 + \frac{0.16}{T} \delta(f)$$

exists at

$$f = 0$$

Clock recovery:

Strong spectral components occur at

$$f = \frac{n}{T}$$

especially at

$$f = \frac{1}{T}$$

since the RZ waveform contains regular transitions. Hence timing recovery can be achieved using PLL/clock recovery circuits.

Example 2

A sequence $\{I_n\}$ consists of IID binary digits selected from the set $\{0, 1\}$. The transmit excitation sequence is

$$B_n = I_n + I_{n-1}$$

The transmitted signal is given by

$$v(t) = \sum_{n=-\infty}^{+\infty} B_n g(t - nT)$$

Draw the constellation diagram and the pmf of B_n .

Assuming

$$g(t) = \cos\left(\frac{\pi t}{T}\right)$$

over

$$[-0.5T, 0.5T]$$

find the transmit power spectral density. Comment on DC content and clock frequency extraction.

Solution:

$$g(t) = \cos\left(\frac{\pi t}{T}\right) \quad \text{over } [-0.5T, 0.5T]$$

$$g(t) = \cos\left(\frac{\pi t}{T}\right) \text{Rect}\left(\frac{t}{T}\right)$$

Fourier Transform of $g(t)$

Using

$$\cos\left(\frac{\pi t}{T}\right) = \frac{1}{2} (e^{j\pi t/T} + e^{-j\pi t/T})$$

$$G(f) = \frac{T}{2} \text{sinc}\left[\left(f - \frac{1}{2T}\right)T\right] + \frac{T}{2} \text{sinc}\left[\left(f + \frac{1}{2T}\right)T\right]$$

Hence,

$$|G(f)|^2 = \frac{T^2}{4} \left[\text{sinc}\left(\left(f - \frac{1}{2T}\right)T\right) + \text{sinc}\left(\left(f + \frac{1}{2T}\right)T\right) \right]^2$$

Using trigonometric simplification,

$$|G(f)|^2 = T^2 \text{sinc}^2(fT) \cos^2\left(\frac{\pi fT}{2}\right)$$

Statistics of I_n

$$\{I_n\} = \{0, 1\} \quad \text{IID}$$

$$P(0) = P(1) = \frac{1}{2}$$

Excitation Sequence

$$B_n = I_n + I_{n-1}$$

Possible values are

$$0 + 0 = 0 \quad P = \frac{1}{4}$$

$$0 + 1 = 1 \quad P = \frac{1}{4}$$

$$1 + 0 = 1 \quad P = \frac{1}{4}$$

$$1 + 1 = 2 \quad P = \frac{1}{4}$$

Hence,

$$P(B_n = 0) = \frac{1}{4}$$

$$P(B_n = 1) = \frac{1}{2}$$

$$P(B_n = 2) = \frac{1}{4}$$

Constellation Diagram

The constellation points are

$$\boxed{\{0, 1, 2\}}$$

on a one-dimensional signal axis.

Autocorrelation Sequence

The PSD is

$$S_v(f) = \frac{|G(f)|^2}{T} \sum_{k=-\infty}^{\infty} R(k) e^{-j2\pi f k T}$$

where

$$R(k) = E[B_n B_{n+k}]$$

Now,

$$B_n = I_n + I_{n-1}$$

Therefore,

$$R(k) = E[(I_n + I_{n-1})(I_{n+k} + I_{n+k-1})]$$

Expanding,

$$R(k) = E[I_n I_{n+k}] + E[I_n I_{n+k-1}] + E[I_{n-1} I_{n+k}] + E[I_{n-1} I_{n+k-1}]$$

Now,

$$E[I_n] = 0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} = \frac{1}{2}$$

Also,

$$E[I_n^2] = 0^2 \cdot \frac{1}{2} + 1^2 \cdot \frac{1}{2} = \frac{1}{2}$$

For $k = 0$,

$$R(0) = E[I_n^2] + E[I_n I_{n-1}] + E[I_{n-1} I_n] + E[I_{n-1}^2]$$

Since IID,

$$E[I_n I_{n-1}] = E[I_n] E[I_{n-1}] = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

Hence,

$$R(0) = \frac{1}{2} + \frac{1}{4} + \frac{1}{4} + \frac{1}{2}$$

$$R(0) = \frac{3}{2}$$

For $k = 1$,

$$R(1) = E[I_n I_{n+1}] + E[I_n^2] + E[I_{n-1} I_{n+1}] + E[I_{n-1} I_n]$$

$$= \frac{1}{4} + \frac{1}{2} + \frac{1}{4} + \frac{1}{4}$$

$$R(1) = \frac{5}{4}$$

Similarly,

$$R(-1) = \frac{5}{4}$$

For $|k| > 1$,

all variables are independent.

Hence,

$$R(k) = 4 \left(\frac{1}{2} \cdot \frac{1}{2} \right)$$

$$R(k) = 1 \quad |k| > 1$$

Thus,

$$R(k) = \begin{cases} \frac{3}{2}, & k = 0 \\ \frac{5}{4}, & k = \pm 1 \\ 1, & |k| > 1 \end{cases}$$

Transmit PSD

$$S_v(f) = \frac{|G(f)|^2}{T} \sum_{k=-\infty}^{\infty} R(k) e^{-j2\pi f k T}$$

Substituting the correlation values,

$$S_v(f) = \frac{|G(f)|^2}{T} \left[\frac{3}{2} + \frac{5}{4} e^{-j2\pi f T} + \frac{5}{4} e^{j2\pi f T} + \sum_{|k|>1} e^{-j2\pi f k T} \right]$$

Using

$$e^{j\theta} + e^{-j\theta} = 2 \cos \theta$$

$$S_v(f) = \frac{|G(f)|^2}{T} \left[\frac{3}{2} + \frac{5}{2} \cos(2\pi f T) + \sum_{|k|>1} e^{-j2\pi f k T} \right]$$

Now,

$$E[B_n] = E[I_n] + E[I_{n-1}] = \frac{1}{2} + \frac{1}{2} = 1$$

Hence, the signal has non-zero mean value and therefore contains spectral line components.

Using

$$\sum_{k=-\infty}^{\infty} e^{-j2\pi f k T} = \frac{1}{T} \sum_{m=-\infty}^{\infty} \delta\left(f - \frac{m}{T}\right)$$

the PSD contains impulse components at

$$f = \frac{m}{T}$$

Therefore,

$$S_v(f) = T \operatorname{sinc}^2(fT) \cos^2\left(\frac{\pi f T}{2}\right) \left[\frac{1}{2} + \frac{5}{2} \cos(2\pi f T) \right] + \text{spectral lines at } f = \frac{m}{T}$$

Comments on DC Content and Clock Recovery

Since

$$E[B_n] = 1$$

the transmitted waveform has a DC component. Hence, a spectral line exists at

$$f = 0$$

The pulse shape

$$g(t) = \cos\left(\frac{\pi t}{T}\right)$$

produces strong spectral energy near

$$f = \frac{1}{2T}$$

Therefore, clock extraction can be achieved using a PLL or narrow band filter centered around

$$f = \frac{1}{2T}$$

Example 3:

Consider the 16-QAM signal constellation shown in the figure below. The distance between any two adjacent points is 'A'. The low pass equivalent of the transmitted signal is

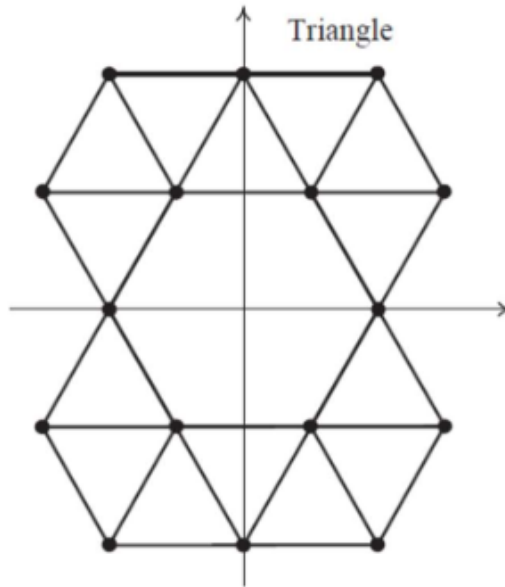


Figure 1: QAM constellation

represented as

$$y(t) = \sum a_n g(t - nT)$$

where ' a_n ' are IID and picked from the constellation. The shaping pulse $g(t)$ is shown below. Find the average energy of the constellation and obtain an expression for the Transmit spectral density.

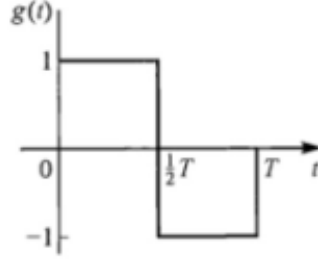


Figure 2: $g(t)$

Solution:

The pulse shape is

$$g(t) = \text{Rect}\left(\frac{t - \frac{T}{4}}{T/2}\right) - \text{Rect}\left(\frac{t - \frac{3T}{4}}{T/2}\right)$$

The transmitted signal is

$$y(t) = \sum_{n=-\infty}^{\infty} a_n g(t - nT)$$

where the data symbols are i.i.d. random variables with

$$a_n = \pm 1$$

with equal probability.

Fourier Transform of the Pulse

Using the Fourier transform pair

$$\text{Rect}\left(\frac{t - t_0}{\tau}\right) \mathcal{F} \tau \text{Sinc}(f\tau) e^{-j2\pi f t_0}$$

where

$$\text{Sinc}(x) = \frac{\sin(\pi x)}{\pi x}$$

For the first rectangle,

$$g_1(t) = \text{Rect}\left(\frac{t - T/4}{T/2}\right)$$

Hence,

$$G_1(f) = \frac{T}{2} \text{Sinc}\left(\frac{fT}{2}\right) e^{-j\omega T/4}$$

where

$$\omega = 2\pi f$$

For the second rectangle,

$$g_2(t) = \text{Rect}\left(\frac{t - 3T/4}{T/2}\right)$$

Hence,

$$G_2(f) = \frac{T}{2} \text{Sinc}\left(\frac{fT}{2}\right) e^{-j3\omega T/4}$$

Since the second pulse is negative,

$$G(f) = G_1(f) - G_2(f)$$

Therefore,

$$G(f) = \frac{T}{2} \text{Sinc}\left(\frac{fT}{2}\right) [e^{-j\omega T/4} - e^{-j3\omega T/4}]$$

Simplification of $G(f)$

Factor out $e^{-j\omega T/2}$:

$$G(f) = \frac{T}{2} \text{Sinc}\left(\frac{fT}{2}\right) e^{-j\omega T/2} [e^{j\omega T/4} - e^{-j\omega T/4}]$$

Using the identity

$$e^{j\theta} - e^{-j\theta} = 2j \sin \theta$$

we obtain

$$G(f) = jT \text{Sinc}\left(\frac{fT}{2}\right) e^{-j\omega T/2} \sin\left(\frac{\omega T}{4}\right)$$

Magnitude Square Spectrum

The magnitude square is

$$|G(f)|^2 = G(f)G^*(f)$$

Thus,

$$|G(f)|^2 = T^2 \text{Sinc}^2\left(\frac{fT}{2}\right) \sin^2\left(\frac{\omega T}{4}\right)$$

Since

$$\omega = 2\pi f$$

we get

$$\frac{\omega T}{4} = \frac{\pi f T}{2}$$

Hence,

$$|G(f)|^2 = T^2 \text{Sinc}^2\left(\frac{fT}{2}\right) \sin^2\left(\frac{\pi f T}{2}\right)$$

Autocorrelation of Data Symbols

The discrete autocorrelation sequence is

$$R_k = E[a_n a_{n+k}]$$

Since the symbols are i.i.d.,

$$E[a_n] = 0$$

For $k \neq 0$,

$$R_k = E[a_n]E[a_{n+k}] = 0$$

For $k = 0$,

$$R_0 = E[a_n^2]$$

Since

$$a_n = \pm 1$$

we have

$$a_n^2 = 1$$

Therefore,

$$R_0 = 1$$

Hence,

$$R_k = \begin{cases} 1, & k = 0 \\ 0, & k \neq 0 \end{cases}$$

Power Spectral Density

The PSD of a PAM signal is

$$S_y(f) = \frac{|G(f)|^2}{T} \sum_{k=-\infty}^{\infty} R_k e^{-j\omega kT}$$

Since

$$R_k = \delta[k]$$

the summation becomes

$$\sum_{k=-\infty}^{\infty} R_k e^{-j\omega kT} = 1$$

Therefore,

$$S_y(f) = \frac{|G(f)|^2}{T}$$

Substituting $|G(f)|^2$,

$$S_y(f) = \frac{1}{T} \left[T^2 \text{Sinc}^2 \left(\frac{fT}{2} \right) \sin^2 \left(\frac{\pi fT}{2} \right) \right]$$

Hence, the final PSD is

$$S_y(f) = T \text{Sinc}^2 \left(\frac{fT}{2} \right) \sin^2 \left(\frac{\pi fT}{2} \right)$$

Important Properties

1. At $f = 0$,

$$S_y(0) = 0$$

Hence the signal has no DC component.

2. The PSD contains spectral nulls at

$$f = \frac{2n}{T}$$

where $n \in \mathbb{Z}$.

3. Due to frequent transitions in the waveform, clock recovery is easier.

Example 4: PSD with Uneven PMF with AMI

Solution:

Given,

$$P(1) = 0.6$$

$$P(0) = 0.4$$

For RZ-AMI coding,

$$a_k \in \{0, +1, -1\}$$

where alternate 1's change polarity.

The Power Spectral Density is

$$S_y(f) = \frac{|G(f)|^2}{T} \sum_{k=-\infty}^{\infty} R_k e^{-j2\pi f k T}$$

where

$$R_k = E[a_n a_{n+k}]$$

is the autocorrelation sequence.

Calculation of R_0

For $k = 0$,

$$R_0 = E[a_n^2]$$

Possible amplitudes are

$$0, \quad +1, \quad -1$$

with probabilities

$$0.4, \quad 0.3, \quad 0.3$$

respectively.

Hence,

$$R_0 = (0)^2(0.4) + (+1)^2(0.3) + (-1)^2(0.3)$$

$$= 0 + 0.3 + 0.3$$

$$\boxed{R_0 = 0.6}$$

Calculation of $R_{\pm 1}$

For $k = \pm 1$,

consider all possible adjacent pairs:

$$(0, 0), \quad (0, 1), \quad (1, 0), \quad (1, 1)$$

Case 1: $(0, 0)$

$$(0)(0)P(0)P(0)$$

$$= (0)(0)(0.4)(0.4) = 0$$

Case 2: $(0, 1)$

$$(0)(1)P(0)P(1)$$

$$= (0)(1)(0.4)(0.6) = 0$$

Case 3: $(1, 0)$

$$(1)(0)P(1)P(0)$$

$$= (1)(0)(0.6)(0.4) = 0$$

Case 4: (1, 1)

Probability of occurrence:

$$P(11) = P(1)P(1)$$

$$= 0.6 \times 0.6$$

$$= 0.36$$

In AMI coding, successive 1's alternate polarity.
Hence,

$$(+1)(-1) = -1$$

or

$$(-1)(+1) = -1$$

Therefore,

$$R_1 = (-1)(0.36)$$

$$\boxed{R_1 = -0.36}$$

Similarly,

$$\boxed{R_{-1} = -0.36}$$

For higher values,

$$R_k = 0, \quad |k| > 1$$

Substituting into PSD Equation

$$S_y(f) = \frac{|G(f)|^2}{T} [R_0 + R_1 e^{-j2\pi fT} + R_{-1} e^{j2\pi fT}]$$

Substituting values,

$$= \frac{|G(f)|^2}{T} [0.6 - 0.36 e^{-j2\pi fT} - 0.36 e^{j2\pi fT}]$$

Taking common factor,

$$= \frac{|G(f)|^2}{T} [0.6 - 0.36 (e^{-j2\pi fT} + e^{j2\pi fT})]$$

Using Euler identity,

$$e^{j\theta} + e^{-j\theta} = 2 \cos \theta$$

Therefore,

$$S_y(f) = \frac{|G(f)|^2}{T} [0.6 - 0.72 \cos(2\pi fT)]$$

Hence,

$$S_y(f) = \frac{|G(f)|^2}{T} [0.6 - 0.72 \cos(2\pi fT)]$$

For RZ Pulse

For Return-to-Zero signaling,

$$g(t) = \begin{cases} 1, & 0 \leq t \leq \frac{T}{2} \\ 0, & \text{otherwise} \end{cases}$$

The Fourier transform is

$$G(f) = \frac{T}{2} \operatorname{sinc}\left(\frac{fT}{2}\right) e^{-j\pi fT/2}$$

Hence,

$$|G(f)|^2 = \frac{T^2}{4} \operatorname{sinc}^2\left(\frac{fT}{2}\right)$$

Finally,

$$S_y(f) = \frac{T}{4} \operatorname{sinc}^2\left(\frac{fT}{2}\right) [0.6 - 0.72 \cos(2\pi fT)]$$

Example 5: PSD of 8-QAM Signal

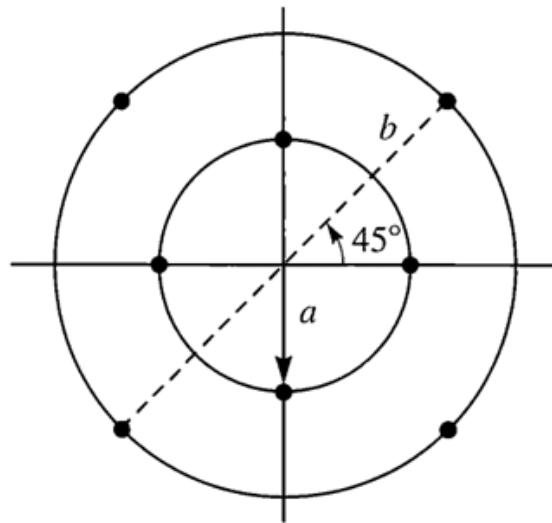


Figure 3: 8-QAM

Solution:

The transmitted signal is

$$x(t) = \sum_{n=-\infty}^{\infty} a_n g(t - nT)$$

where

- a_n are i.i.d. random symbols
- $g(t)$ is the pulse shape
- T is the symbol duration

The 8-QAM constellation consists of:

$$\pm r_1, \quad \pm jr_1$$

and

$$\pm r_2 e^{j\theta}, \quad \pm r_2 e^{-j\theta}$$

All symbols are equally likely.

Hence,

$$P(a_n) = \frac{1}{8}$$

Pulse Shape

Given,

$$g(t) = \text{Rect}\left(\frac{t}{T}\right)$$

The Fourier transform of rectangular pulse is

$$G(f) = T \text{sinc}(fT) e^{-j\pi fT}$$

Therefore,

$$|G(f)|^2 = T^2 \text{sinc}^2(fT)$$

PSD Formula

For linear modulation,

$$S_x(f) = \frac{|G(f)|^2}{T} \sum_{k=-\infty}^{\infty} R(k) e^{-j2\pi f k T}$$

where

$$R(k) = E[a_n a_{n+k}^*]$$

is the autocorrelation sequence.

Calculation of Mean

The constellation points are symmetric about the origin.

Hence,

$$E[a_n] = \frac{1}{8} \left[r_1 - r_1 + jr_1 - jr_1 \right. \\ \left. + r_2 e^{j\theta} - r_2 e^{j\theta} + r_2 e^{-j\theta} - r_2 e^{-j\theta} \right]$$

All terms cancel.

Therefore,

$$\boxed{E[a_n] = 0}$$

Calculation of $R(0)$

By definition,

$$R(0) = E[a_n a_n^*]$$

Since

$$a_n a_n^* = |a_n|^2$$

we get

$$R(0) = E[|a_n|^2]$$

Calculation of $R(1)$

By definition,

$$R(1) = E[a_n a_{n+1}^*]$$

Since the symbols are independent,

$$R(1) = E[a_n] E[a_{n+1}^*]$$

But,

$$E[a_n] = 0$$

and

$$E[a_{n+1}^*] = 0$$

Hence,

$$R(1) = 0 \times 0$$

Therefore,

$$\boxed{R(1) = 0}$$

Similarly,

$$\boxed{R(k) = 0, \quad k \neq 0}$$

Energy of Inner Constellation Points

The four inner points are

$$+r_1, \quad -r_1, \quad +jr_1, \quad -jr_1$$

Each point has energy

$$|a_n|^2 = r_1^2$$

Contribution:

$$4 \left(\frac{1}{8} \right) r_1^2 = \frac{1}{2} r_1^2$$

Energy of Outer Constellation Points

The four outer points are

$$\pm r_2 e^{j\theta}, \quad \pm r_2 e^{-j\theta}$$

Each point has energy

$$|r_2 e^{j\theta}|^2 = r_2^2$$

since

$$|e^{j\theta}| = 1$$

Contribution:

$$4 \left(\frac{1}{8} \right) r_2^2 = \frac{1}{2} r_2^2$$

Total Average Energy

Therefore,

$$R(0) = \frac{1}{2} r_1^2 + \frac{1}{2} r_2^2$$

Hence,

$$\boxed{R(0) = \frac{r_1^2 + r_2^2}{2}}$$

Substituting into PSD

Since only $R(0)$ is nonzero,

$$S_x(f) = \frac{|G(f)|^2}{T} R(0)$$

Substituting $R(0)$,

$$S_x(f) = \frac{|G(f)|^2}{T} \cdot \frac{r_1^2 + r_2^2}{2}$$

Therefore,

$$S_x(f) = \frac{|G(f)|^2}{2T}(r_1^2 + r_2^2)$$

Now substitute

$$|G(f)|^2 = T^2 \text{sinc}^2(fT)$$

Hence,

$$S_x(f) = \frac{T^2}{2T}(r_1^2 + r_2^2) \text{sinc}^2(fT)$$

Finally,

$$S_x(f) = \frac{T}{2}(r_1^2 + r_2^2) \text{sinc}^2(fT)$$

Example 6

The transmitted signal is

$$y(t) = \sum_{n=-\infty}^{\infty} B_n g(t - nT)$$

where

$$B_n = I_n + I_{n-1}$$

and

$$I_n \in \{+1, -1\}$$

with equal probability,

$$P(I_n = 1) = P(I_n = -1) = \frac{1}{2}$$

The pulse shape is

$$g(t) = \text{Rect}\left(\frac{t}{T}\right)$$

with

$$T = \frac{1}{2W}$$

PMF of B_n

Possible values of B_n are obtained as follows:

I_n	I_{n-1}	B_n
1	1	2
1	-1	0
-1	1	0
-1	-1	-2

Each combination occurs with probability

$$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

Hence,

$$P(B_n = 2) = \frac{1}{4}$$

$$P(B_n = 0) = \frac{1}{2}$$

$$P(B_n = -2) = \frac{1}{4}$$

PSD Formula

The PSD is

$$S_y(f) = \frac{|G(f)|^2}{T} \sum_{k=-\infty}^{\infty} R(k) e^{j\omega k T}$$

where

$$R(k) = E[B_n B_{n+k}]$$

is the autocorrelation sequence.

Calculation of $R(0)$

By definition,

$$R(0) = E[B_n^2]$$

Substituting

$$B_n = I_n + I_{n-1}$$

we get

$$R(0) = E[(I_n + I_{n-1})^2]$$

Expanding,

$$R(0) = E[I_n^2] + E[I_{n-1}^2] + 2E[I_n I_{n-1}]$$

Since I_n and I_{n-1} are independent,

$$E[I_n I_{n-1}] = E[I_n]E[I_{n-1}]$$

Now,

$$E[I_n] = 1 \left(\frac{1}{2} \right) + (-1) \left(\frac{1}{2} \right) = 0$$

Hence,

$$E[I_n I_{n-1}] = 0$$

Also,

$$E[I_n^2] = 1^2 \left(\frac{1}{2} \right) + (-1)^2 \left(\frac{1}{2} \right) = 1$$

Similarly,

$$E[I_{n-1}^2] = 1$$

Therefore,

$$R(0) = 1 + 1 + 0$$

Hence,

$$\boxed{R(0) = 2}$$

Calculation of $R(1)$

By definition,

$$R(1) = E[B_n B_{n+1}]$$

Now,

$$B_n = I_n + I_{n-1}$$

and

$$B_{n+1} = I_{n+1} + I_n$$

Thus,

$$R(1) = E[(I_n + I_{n-1})(I_{n+1} + I_n)]$$

Expanding,

$$R(1) = E[I_n I_{n+1}] + E[I_n^2] + E[I_{n-1} I_{n+1}] + E[I_{n-1} I_n]$$

Since all samples are independent,

$$E[I_n I_{n+1}] = 0$$

$$E[I_{n-1} I_{n+1}] = 0$$

$$E[I_{n-1}I_n] = 0$$

Only one term remains:

$$E[I_n^2] = 1$$

Therefore,

$$\boxed{R(1) = 1}$$

Calculation of $R(-1)$

Similarly,

$$R(-1) = E[B_n B_{n-1}]$$

where

$$B_{n-1} = I_{n-1} + I_{n-2}$$

Thus,

$$R(-1) = E[(I_n + I_{n-1})(I_{n-1} + I_{n-2})]$$

Expanding,

$$R(-1) = E[I_n I_{n-1}] + E[I_n I_{n-2}] + E[I_{n-1}^2] + E[I_{n-1} I_{n-2}]$$

All cross terms are zero.

Hence,

$$R(-1) = E[I_{n-1}^2]$$

$$\boxed{R(-1) = 1}$$

Calculation of $R(k)$, $|k| \geq 2$

For $|k| \geq 2$, there are no common random variables.

Hence all terms become zero.

Therefore,

$$\boxed{R(k) = 0, \quad |k| \geq 2}$$

Substituting into PSD

Substituting the autocorrelation values,

$$S_y(f) = \frac{|G(f)|^2}{T} [R(0) + R(1)e^{j\omega T} + R(-1)e^{-j\omega T}]$$

Substituting values,

$$S_y(f) = \frac{|G(f)|^2}{T} [2 + e^{j\omega T} + e^{-j\omega T}]$$

Using Euler identity,

$$e^{j\theta} + e^{-j\theta} = 2 \cos \theta$$

we get

$$S_y(f) = \frac{|G(f)|^2}{T} [2 + 2 \cos(\omega T)]$$

Fourier Transform of Rectangular Pulse

For

$$g(t) = \text{Rect}\left(\frac{t}{T}\right)$$

the Fourier transform is

$$G(f) = T \text{sinc}(fT) e^{-j\pi fT}$$

Hence,

$$|G(f)|^2 = T^2 \text{sinc}^2(fT)$$

Substituting into PSD,

$$S_y(f) = \frac{T^2 \text{sinc}^2(fT)}{T} [2 + 2 \cos(\omega T)]$$

Therefore,

$$S_y(f) = T \text{sinc}^2(fT) [2 + 2 \cos(\omega T)]$$

Taking common factor,

$$S_y(f) = 2T \text{sinc}^2(fT) [1 + \cos(\omega T)]$$

Using identity

$$1 + \cos \theta = 2 \cos^2\left(\frac{\theta}{2}\right)$$

Final PSD expression becomes

$$S_y(f) = 4T \text{sinc}^2(fT) \cos^2\left(\frac{\omega T}{2}\right)$$

Example 7

Consider the random sequence

$$B_n = I_n + I_{n-1}$$

where

$$I_n \in \{0, 1\}$$

with

$$P(I_n = 0) = P(I_n = 1) = \frac{1}{2}$$

and $\{I_n\}$ is an i.i.d. sequence.

Determine the autocorrelation function $R_B(k)$ and obtain the PSD.

Solution

Given,

$$B_n = I_n + I_{n-1}$$

Possible values of B_n are

I_n	I_{n-1}	B_n
0	0	0
0	1	1
1	0	1
1	1	2

Hence,

$$P(B_n = 0) = \frac{1}{4}$$

$$P(B_n = 1) = \frac{1}{2}$$

$$P(B_n = 2) = \frac{1}{4}$$

The autocorrelation is

$$R_B(k) = E[B_n B_{n+k}]$$

Substituting B_n ,

$$\begin{aligned} R_B(k) &= E[(I_n + I_{n-1})(I_{n+k} + I_{n+k-1})] \\ &= E[I_n I_{n+k}] + E[I_n I_{n+k-1}] \\ &\quad + E[I_{n-1} I_{n+k}] + E[I_{n-1} I_{n+k-1}] \end{aligned}$$

Since $I_n \in \{0, 1\}$,

$$E[I_n] = 0 \left(\frac{1}{2}\right) + 1 \left(\frac{1}{2}\right) = \frac{1}{2}$$

For $k \neq 0$,

$$E[I_n I_{n+k}] = E[I_n] E[I_{n+k}] = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

For $k = 0$,

$$E[I_n^2] = 0^2 \left(\frac{1}{2}\right) + 1^2 \left(\frac{1}{2}\right) = \frac{1}{2}$$

Therefore,

$$E[I_n I_m] = \begin{cases} \frac{1}{2}, & n = m \\ \frac{1}{4}, & n \neq m \end{cases}$$

Case 1: $k = 0$

$$\begin{aligned} R_B(0) &= E[I_n^2] + E[I_n I_{n-1}] \\ &\quad + E[I_{n-1} I_n] + E[I_{n-1}^2] \\ &= \frac{1}{2} + \frac{1}{4} + \frac{1}{4} + \frac{1}{2} \\ &= 1.5 \end{aligned}$$

Hence,

$$R_B(0) = 1.5$$

Case 2: $k = 1$

$$\begin{aligned} R_B(1) &= E[I_n I_{n+1}] + E[I_n I_n] \\ &\quad + E[I_{n-1} I_{n+1}] + E[I_{n-1} I_n] \\ &= \frac{1}{4} + \frac{1}{2} + \frac{1}{4} + \frac{1}{4} \\ &= 1.25 \end{aligned}$$

Thus,

$$R_B(1) = 1.25$$

Similarly,

$$R_B(-1) = 1.25$$

Case 3: $k = 2$

$$\begin{aligned} R_B(2) &= E[I_n I_{n+2}] + E[I_n I_{n+1}] \\ &\quad + E[I_{n-1} I_{n+2}] + E[I_{n-1} I_{n+1}] \\ &= \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} \\ &= 1 \end{aligned}$$

Hence,

$$R_B(2) = 1$$

Similarly,

$$R_B(-2) = 1$$

and for $|k| \geq 2$,

$$R_B(k) = 1$$

Therefore,

$$R_B(k) = \begin{cases} 1.5, & k = 0 \\ 1.25, & k = \pm 1 \\ 1, & |k| \geq 2 \end{cases}$$

PSD

The PSD is

$$S_B(\omega) = \sum_{k=-\infty}^{\infty} R_B(k) e^{-j\omega k}$$

Substituting the autocorrelation values,

$$S_B(\omega) = 1.5 + 1.25e^{j\omega} + 1.25e^{-j\omega} + e^{j2\omega} + e^{-j2\omega} + \dots$$

Using

$$e^{j\omega} + e^{-j\omega} = 2 \cos \omega$$

we obtain

$$S_B(\omega) = 1.5 + 2(1.25) \cos \omega + 2 \cos 2\omega + \dots$$

Including the pulse shaping function $G(f)$,

$$S_x(\omega) = \frac{|G(f)|^2}{T} [1.5 + 2(1.25) \cos(\omega T) + 2 \cos(2\omega T) + \dots]$$